How Drug Characteristics Affect User Satisfaction

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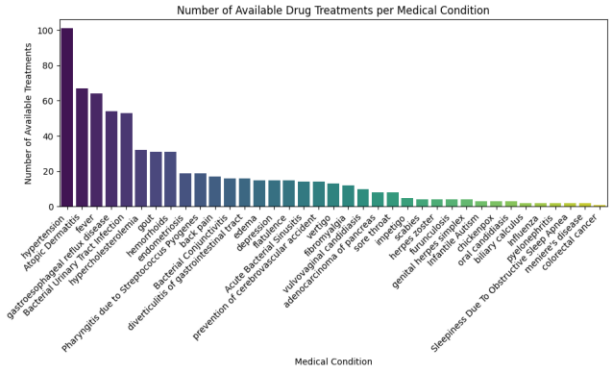
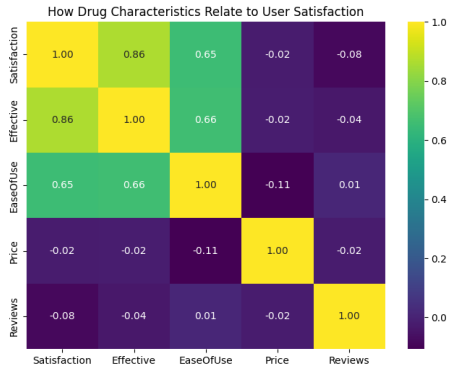
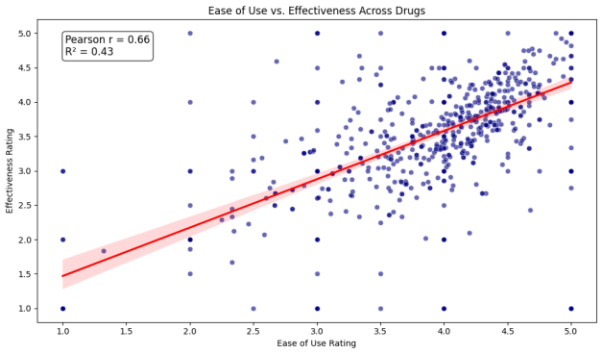
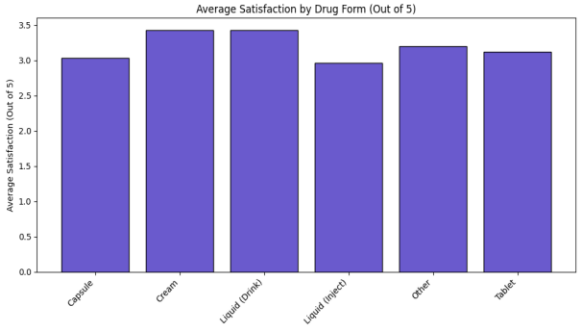
Project Track: Data Science and Analytics
Major: Statistics

Problem Statement : This project examines what affects user satisfaction with drugs, including factors such as effectiveness, ease of use, price, form, and the medical condition treated. Satisfaction matters because it influences how well patients follow treatments and how they feel about their care. By examining user ratings alongside drug availability, these insights identify the most influential features and support more informed choices for patients, clinicians, and pharmaceutical developers.









Method:

Dataset: The dataset, from Kaggle's *Drug Performance Evaluation*, contains information on drug characteristics and user satisfaction, allowing analysis of which features influence satisfaction.
Data Visualization: I created the dashboard using Python with Matplotlib and Seaborn, cleaning the data and selecting the necessary columns. Bar charts show satisfaction by form and the number of available treatments for each condition, heatmaps display feature correlations, and a scatterplot examines ease of use versus effectiveness, highlighting the factors that most influence user satisfaction.

Results:

The analysis showed that effectiveness and ease of use are the main factors affecting user satisfaction, with easier-to-use drugs generally rated as more effective. There is a moderate to strong positive relationship between ease of use and perceived effectiveness, with a Pearson correlation of 0.66 and an R^2 of 0.43. The drug form also affects satisfaction, with drinkable liquid and cream forms rated higher and injectable liquids lower. Drug availability varies by condition, helping patients choose medicines, guiding providers, and supporting drug makers in improving user experience.

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Kaggle Dataset Link: https://www.kaggle.com/datasets/thedevastator/drug-performance-evaluation
Contact: xu.5067@buckeyemail.osu.edu
LinkedIn: https://www.linkedin.com/in/claire-xu-07508a135/